

Weekly Progress Report

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Domain: Core Java Development (Music Player with Playlist System)

Date of submission: 11/05/2026

Week Ending: 04

I. Overview

This week, the primary focus was on developing an Advanced Music Player Application using Core Java, Swing GUI, and JDBC connectivity.

The project aims to provide users with a complete music management system where they can manage songs, create playlists, control music playback, and store playlist data using a database system.

The application was designed with a user-friendly graphical interface and menu-driven functionality for smooth interaction.

II. Achievements

1. Project Understanding and Planning

- Studied the problem statement for the Music Player with Playlist System.
- Identified important modules and functionalities.
- Designed the workflow and application structure.
- Planned database connectivity using JDBC API.

Key Modules Identified:

- Music Playback
- Music Library Management
- Playlist Management
- Search and Filtering
- Database Connectivity
- Error Handling
- GUI Interface

2. Core Java and Swing Implementation

Name of the Project:

Music Player with Playlist Management System

Developed Components:

- Main Music Player Window
- Song List Panel
- Playlist Panel
- Playlist Songs Section
- Playback Control Buttons
- Dialog Boxes and Notifications

Implemented Features:

- Display available songs
- Create playlists
- Add songs to playlists
- Remove songs from playlists
- Play selected songs
- Pause music playback
- Stop playback
- Skip to next track
- Dynamic song selection handling
- Menu-driven navigation

3. JDBC Database Integration

Database Features Implemented:

- Database connection using JDBC API
- Song information storage
- Playlist data management
- Retrieval of song records from database
- Insert and delete operations using SQL queries

Tables Planned/Created:

- Songs Table

- Playlists Table
- Playlist Songs Mapping Table

SQL Operations Practiced:

- INSERT
- SELECT
- DELETE
- UPDATE

4. GUI Development

Designed graphical interface using Java Swing components

Swing Components Used:

- **JFrame**
- **JPanel**
- **JButton**
- **JLabel**
- **JList**
- **JScrollPane**
- **JOptionPane**
- **BorderLayout**
- **GridLayout**

GUI Functionalities:

- Interactive buttons
- Playlist selection
- Song selection
- Playback controls
- Confirmation messages

5. Learning Core Java Concepts

Strengthened understanding of:

- **Object-Oriented Programming (OOP)**
- **Classes and Objects**
- **Constructors and Methods**

- **Event Handling**
- **Java Swing Framework**
- **JDBC Connectivity**
- **Exception Handling**
- **Collections Framework**
- **ArrayList and DefaultListModel**

Improved logical thinking and modular coding practices.

III. Challenges

1. Swing GUI Execution on Online Platforms

Faced issues running Java Swing applications on online compilers due to X11 display server limitations.

Error Encountered:

Can't connect to X11 window server

Solution:

- Converted project into console-based version for testing.
- Used local IDE environments for GUI execution.

2. JDBC Connectivity Issues

Challenges faced while:

- Establishing database connections
- Loading JDBC drivers
- Managing SQL exceptions

Solution:

- Practiced JDBC syntax and connection setup
- Improved exception handling methods

3. Event Handling Complexity

Managing multiple button actions and playlist updates in GUI became complex.

Solution:

- Implemented separate action listeners

- Organized code into logical sections

IV. Learning Resources

1. Java Learning Materials

Referred to:

Java documentation

JDBC tutorials

Swing GUI tutorials

Online programming references

2. Practical Learning

- Tested modules individually
- Debugged application using IDE
- Improved GUI structure step by step
- Practiced SQL query execution

IDEs Used:

- IntelliJ IDEA
- Eclipse IDE
- OnlineGDB

V. Next Week's Goals

1. Feature Enhancement

Planned features:

- Real audio playback integration
- Song duration display
- Volume control
- Shuffle and repeat functionality
- Playlist saving system
- Song reordering within playlists

2. Database Improvements

- Improve SQL query optimization

- Add advanced search filters
- Implement genre-based filtering
- Improve playlist retrieval performance

3. User Interface Improvement

- Better UI design and formatting
- Add icons and images
- Improve navigation experience
- Add music progress bar

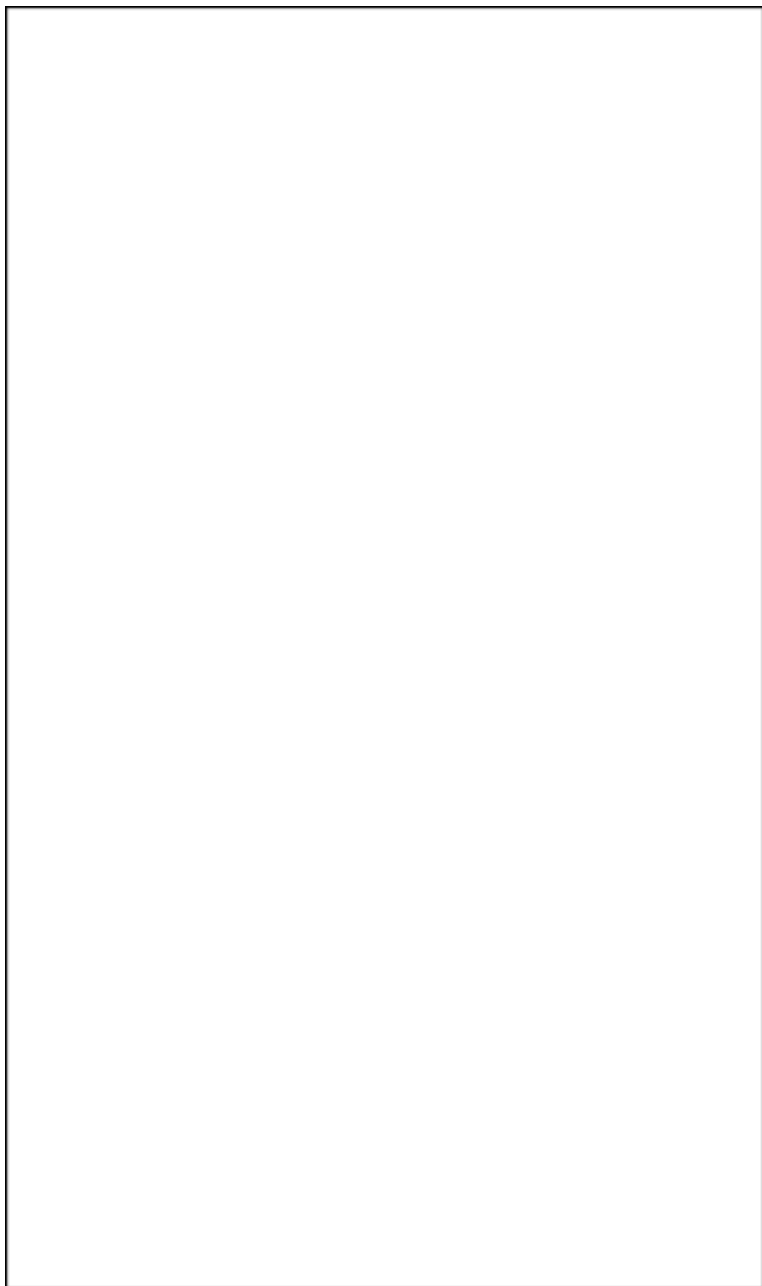
VI. Additional Comments

Project Links :

GitHub

<https://github.com/mantasha20Shaikh-droid/Core-Java-Projects/blob/main/Problem-05%20Advanced%20Project.java>

Project Report :



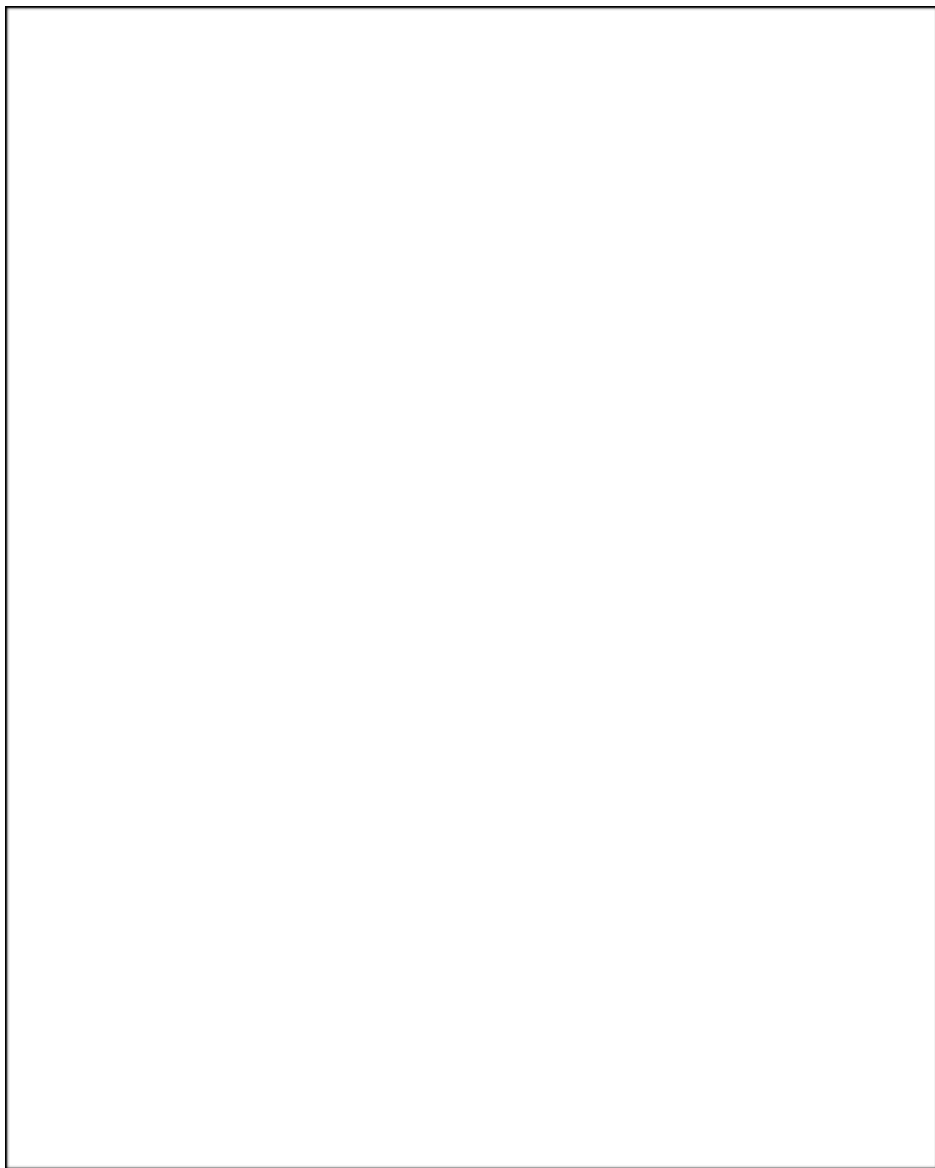
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===== MUSIC PLAYER =====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song from Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Next Song
0. Exit
Enter Choice: 1

--- SONG LIST ---
1. Believer - Imagine Dragons
2. Shape of You - Ed Sheeran
3. Perfect - Ed Sheeran

===== MUSIC PLAYER =====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song from Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Next Song
0. Exit
Enter Choice: 3

Select Song Number:
1. Believer - Imagine Dragons
2. Shape of You - Ed Sheeran
3. Perfect - Ed Sheeran
2
Song Added Successfully
```



```
===== MUSIC PLAYER =====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song from Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Next Song
0. Exit
```

Enter Choice: 4

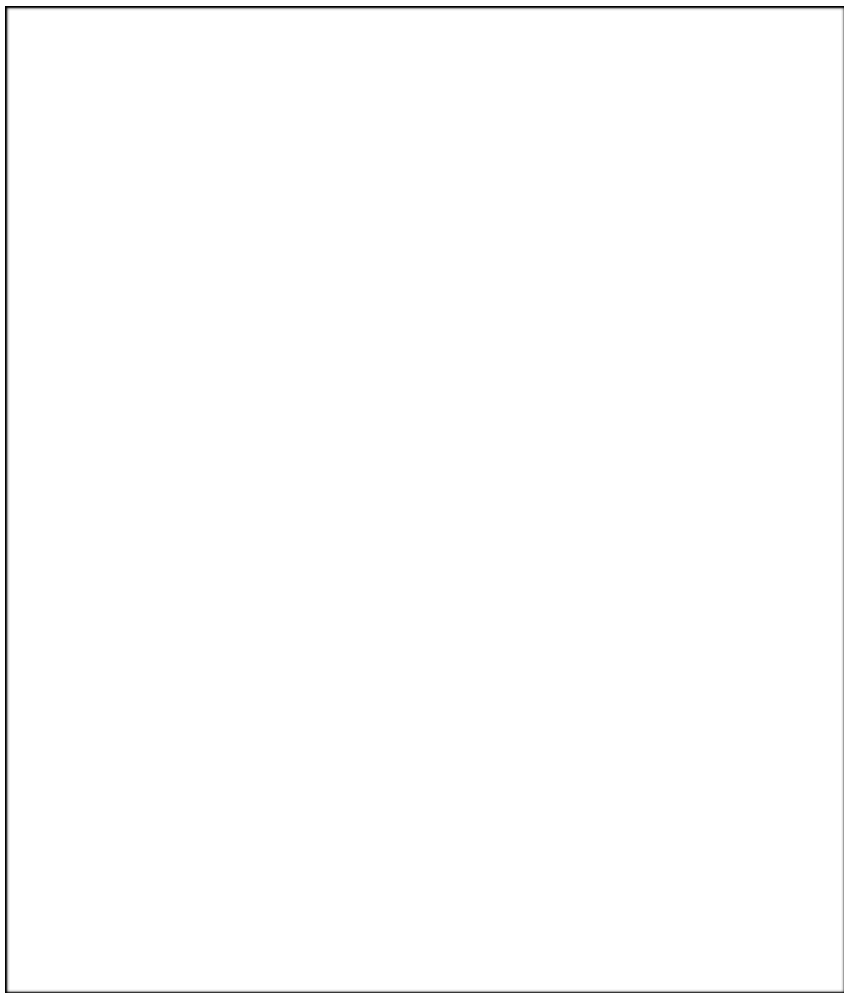
```
--- PLAYLIST SONGS ---
1. Shape of You - Ed Sheeran
```

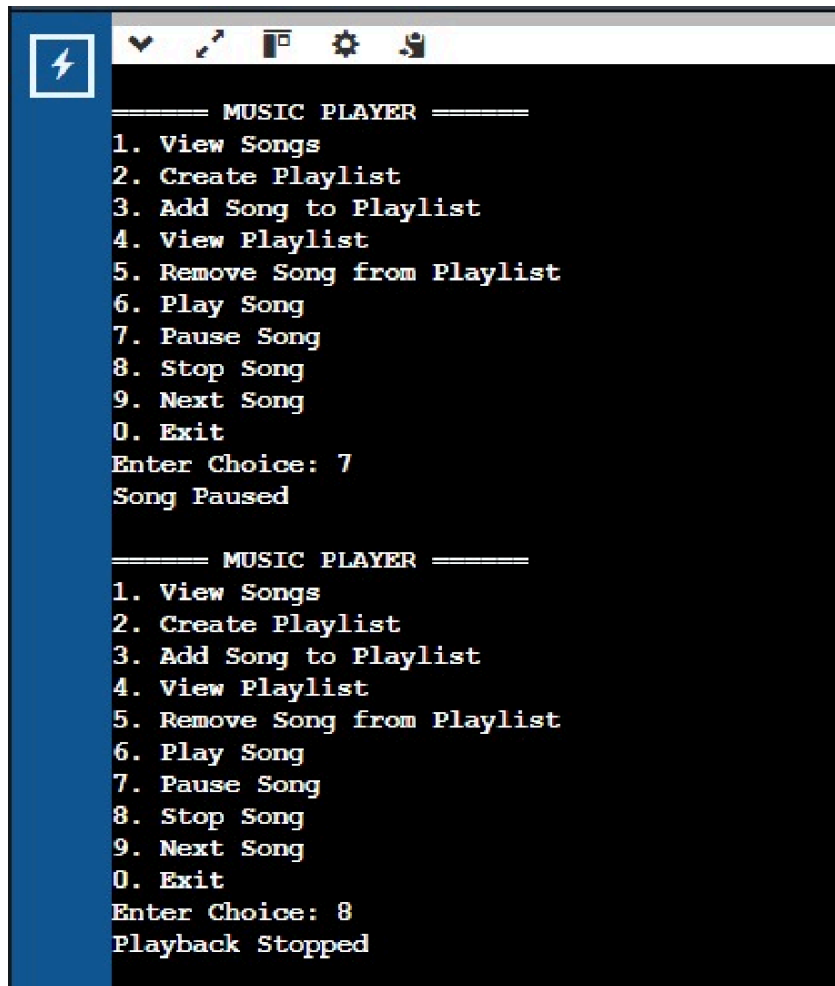
```
===== MUSIC PLAYER =====
1. View Songs
2. Create Playlist
3. Add Song to Playlist
4. View Playlist
5. Remove Song from Playlist
6. Play Song
7. Pause Song
8. Stop Song
9. Next Song
0. Exit
```

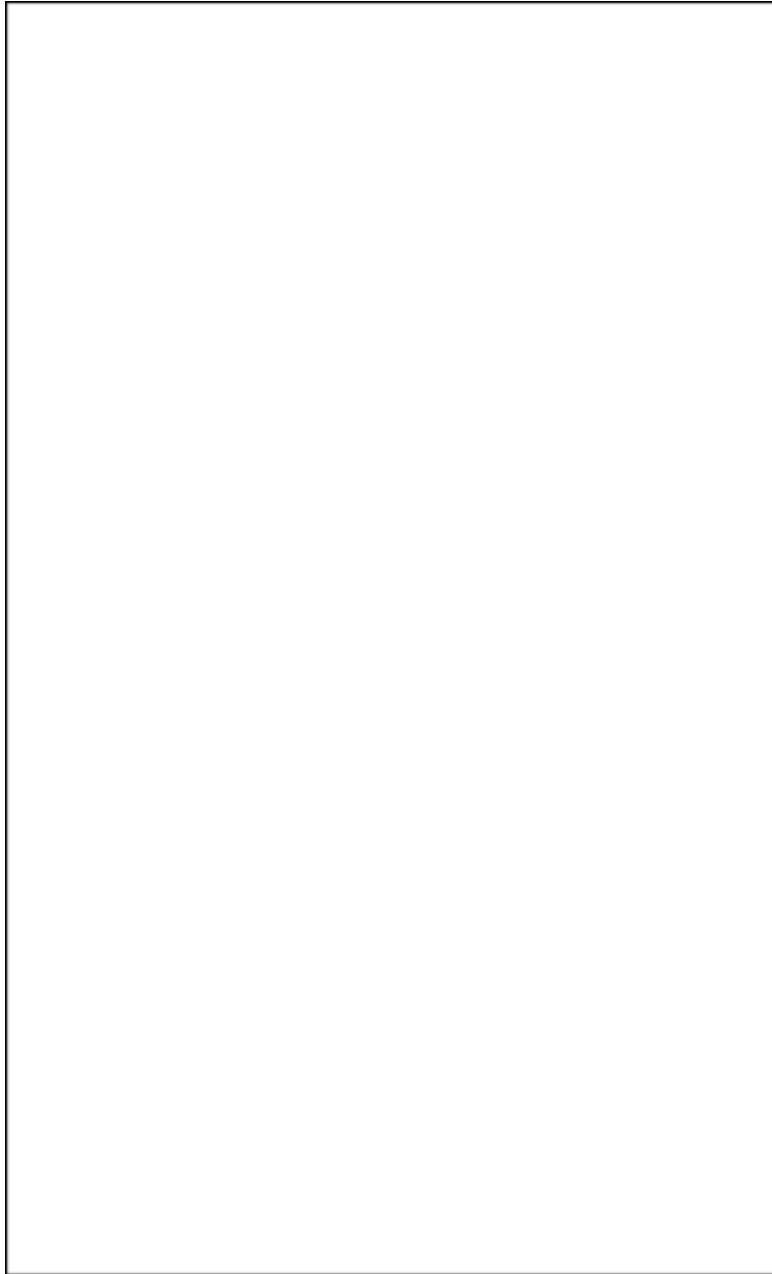
Enter Choice: 6

```
Select Song to Play:
1. Shape of You - Ed Sheeran
1
```

Now Playing: Shape of You - Ed Sheeran









8. Stop Song

9. Next Song

0. Exit

Enter Choice: 9

Playing Next Song

===== MUSIC PLAYER =====

1. View Songs

2. Create Playlist

3. Add Song to Playlist

4. View Playlist

5. Remove Song from Playlist

6. Play Song

7. Pause Song

8. Stop Song

9. Next Song

0. Exit

Enter Choice: 1

---- SONG LIST ----

1. Believer - Imagine Dragons

2. Shape of You - Ed Sheeran

3. Perfect - Ed Sheeran

===== MUSIC PLAYER =====

1. View Songs

2. Create Playlist

3. Add Song to Playlist

4. View Playlist

5. Remove Song from Playlist

6. Play Song

7. Pause Song

8. Stop Song

9. Next Song

0. Exit

Enter Choice: 0

Exiting Music Player...